Remote Sensing Image Classification of Land Use and Land Cover Using Deep Learning

Gowri J, Navadeep P, Jagan T
Department of Computer Science and Engineering
Srinivasa Ramanujan Institute of Technology
Anantapur, India
Under the Guidance of Ms.G.Lokeswari M.Tech., (Ph.D)

Abstract—**Land Use and Land Cover (LULC) classification is one of the most important applications of remote sensing, as it provides critical information for environmental monitoring, urban planning, agriculture management, disaster response, and sustainable development. With the rapid growth of satellite and aerial imaging technologies, large volumes of high-resolution remote sensing data are generated every day. Although these data offer great potential for understanding changes in the Earth's surface, extracting useful and reliable information from them remains a challenging task. Traditional LULC classification approaches, which rely on manual interpretation, pixel-based techniques, and handcrafted features, are no longer sufficient to handle the complexity, diversity, and scale of modern satellite imagery.**

Conventional machine learning methods depend heavily on expert-designed features such as spectral indices, texture descriptors, and shape-based rules. These features often fail to capture the complex spatial relationships and subtle variations present in real-world land cover types. In addition, such methods are sensitive to noise, illumination changes, seasonal variations, and sensor differences, which leads to reduced accuracy and poor generalization across different geographical regions.

To address these challenges, this paper proposes a deep learning–based framework for automated and accurate land use and land cover classification using satellite images. The EuroSAT RGB dataset is used as the benchmark dataset, as it contains diverse land cover classes captured from real satellite imagery. Multiple deep convolutional neural network architectures, including ResNet50, ResNet18, DenseNet121, VGG19, ResNet152, and ConvNeXt-Tiny, are implemented and trained to analyze their ability to learn spatial and spectral patterns from satellite images. These models represent different design philosophies in deep learning, ranging from residual learning and dense connectivity to modern convolutional architectures.

To improve the robustness and generalization capability of the models, extensive image preprocessing is performed. This includes rescaling of pixel values, geometric transformations, horizontal flipping, and reshaping of input images to a uniform size. These operations help the models learn invariant and discriminative features, reducing overfitting and improving performance on unseen data.

The performance of each model is evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. In addition, Grad-CAM (Gradient-weighted Class Activation Mapping) is employed as an Explainable Artificial Intelligence (XAI) technique to visualize the important regions of satellite images that influence the model's predictions. This improves transparency and trust by showing how the deep learning model arrives at its decisions.

Experimental results demonstrate that ConvNeXt-Tiny outperforms the other architectures, achieving an accuracy of 98.49%, precision of 98.48%, recall of 98.46%, and an F1-score of 98.47%. The best-performing model is then integrated into a Flask-based web application with SQLite-based user authentication. This web platform allows users to upload satellite images, perform automated preprocessing, obtain real-time LULC classification results, and visualize Grad-CAM heatmaps. The developed system provides an end-to-end, scalable, and interpretable solution for practical land cover analysis.

***Index Terms*—Land Use and Land Cover (LULC), Remote Sensing, Deep Learning, Convolutional Neural Networks, ConvNeXt, ResNet, DenseNet, VGG, Grad-CAM, Explainable AI, Satellite Image Classification, Flask Web Application.**

I. Introduction

Remote sensing has become one of the most important technologies for observing and understanding the Earth's surface. Using satellites and aerial sensors, remote sensing systems continuously collect images that capture information about vegetation, water bodies, urban areas, agricultural fields, forests, deserts, and many other land cover types. This information is essential for a wide range of applications, including environmental protection, urban development, agricultural planning, climate change monitoring, disaster management, and natural resource management.

Land Use and Land Cover (LULC) refers to both how land is utilized by humans and the physical materials present on the Earth's surface. Land use includes activities such as residential development, industrial areas, farming, and transportation, while land cover refers to natural and man-made features such as vegetation, water, soil, and buildings. Accurate LULC maps are required by governments, urban planners, environmental agencies, and researchers to make informed decisions about land management, infrastructure development, and ecological conservation.

With the advancement of satellite technology, modern remote sensing systems can capture images with very high spatial and spectral resolution. While this provides more detailed information about the Earth's surface, it also creates new challenges. The large volume of data and the complex patterns present in high-resolution images make manual interpretation impractical. Traditional pixel-based classification methods, which analyze each pixel independently based on its spectral value, often fail to capture spatial context and relationships between neighboring regions. As a result, these methods produce noisy and inaccurate classification results, especially in heterogeneous landscapes.

Machine learning techniques were introduced to improve LULC classification by using statistical models trained on

labeled data. Methods such as Support Vector Machines, Random Forests, and k-Nearest Neighbors rely on handcrafted features extracted from images, such as spectral indices, texture measures, and shape descriptors. Although these approaches improved performance compared to simple threshold-based methods, they still depend heavily on expert knowledge for feature design and do not generalize well across different regions and imaging conditions.

Deep learning, particularly Convolutional Neural Networks (CNNs), has emerged as a powerful tool for image analysis. CNNs are capable of automatically learning hierarchical feature representations from raw pixel data. Lower layers of a CNN learn simple features such as edges and corners, while deeper layers capture complex patterns such as textures, shapes, and spatial arrangements. This ability makes CNNs especially suitable for analyzing satellite images, where land cover classes are defined not only by color but also by spatial structure and context.

Recent research has shown that deep CNNs significantly outperform traditional machine learning methods in remote sensing image classification. Architectures such as VGG, ResNet, and DenseNet have been successfully applied to LULC classification tasks. More recently, modern convolutional architectures such as ConvNeXt have been developed, combining the strengths of convolutional networks with design ideas inspired by transformer models. These architectures offer improved accuracy and efficiency, making them well suited for large-scale satellite image analysis.

Despite these advances, many deep learning-based LULC classification systems remain limited to offline experiments and lack practical deployment. Furthermore, deep learning models are often treated as black boxes, making it difficult for users to understand or trust their predictions. There is a growing need for systems that not only provide high classification accuracy but also offer interpretability and real-time accessibility.

In this work, a comprehensive deep learning–based framework for LULC classification is proposed. Multiple CNN architectures are trained and compared using the EuroSAT RGB dataset. The best-performing model is integrated into a Flask-based web application that allows users to upload satellite images and obtain classification results along with Grad-CAM visual explanations. This approach bridges the gap between high-performance deep learning research and practical, user-friendly remote sensing applications.

II. Literature Survey

Land Use and Land Cover (LULC) classification has been a major research topic in remote sensing for several decades because of its importance in environmental analysis, urban planning, agriculture, and climate studies. Early research in this field relied primarily on manual interpretation and pixel-based statistical methods. In these approaches, satellite images were classified by analyzing the spectral values of individual pixels and assigning them to predefined land cover classes. Techniques such as Maximum Likelihood Classification, k-means clustering, and threshold-based rules were commonly used. Although these methods were simple to implement, they were highly sensitive to noise, illumination changes, and mixed pixels, making them unsuitable for complex and high-resolution imagery.

To overcome the limitations of purely statistical methods, machine learning techniques such as Support Vector Machines (SVM), Decision Trees, Random Forests, and k-Nearest Neighbors (k-NN) were introduced for LULC classification. These models used handcrafted features extracted from satellite images, including spectral indices (such as NDVI), texture features, and shape descriptors. Researchers reported improved classification accuracy compared to traditional pixel-based methods. However, these approaches required significant domain expertise for feature engineering and parameter tuning. In addition, their performance often degraded when applied to new geographic regions or images captured under different seasonal and atmospheric conditions.

With the increasing availability of high-resolution satellite imagery, the limitations of handcrafted feature-based methods became more apparent. High-resolution images contain complex spatial structures such as roads, buildings, agricultural patterns, and forest textures, which are difficult to capture using simple statistical features. This led researchers to explore deep learning techniques, particularly Convolutional Neural Networks (CNNs), for remote sensing image classification. CNNs automatically learn multi-level features directly from raw pixel data, making them more robust and adaptive to complex visual patterns.

Several studies have applied CNN architectures such as AlexNet, VGGNet, and GoogLeNet to LULC classification tasks. These models demonstrated significantly higher accuracy compared to traditional machine learning approaches. VGGNet, with its deep and uniform architecture, was found to be effective in learning fine-grained spatial features. GoogLeNet, using inception modules, captured multi-scale information, which is useful for analyzing land cover types of different sizes. However, these early deep learning models were computationally expensive and sometimes difficult to train for large datasets.

Residual Networks (ResNet) introduced shortcut connections that allow information to bypass certain layers, making it possible to train very deep networks without suffering from vanishing gradient problems. ResNet-based models such as ResNet50 and ResNet152 have been widely used in remote sensing applications and have shown strong performance in LULC classification. These networks can learn both low-level and high-level spatial features, enabling better discrimination between visually similar land cover classes such as urban areas and barren land.

DenseNet further improved feature propagation by connecting each layer to all subsequent layers, promoting feature reuse and improving gradient flow. DenseNet121 has been applied in several studies for satellite image classification

and has demonstrated good performance, especially when training data is limited. The dense connectivity helps the network capture fine-grained texture patterns that are important for distinguishing different land cover types.

More recently, modern convolutional architectures such as ConvNeXt have been proposed. ConvNeXt models incorporate design principles inspired by vision transformers while retaining the efficiency of convolutional networks. These models use simplified layer structures, larger receptive fields, and optimized normalization strategies to achieve strong representation learning with fewer parameters. Initial studies have shown that ConvNeXt performs competitively with transformer-based models while being more computationally efficient, making it suitable for large-scale remote sensing applications.

Another important research direction in LULC classification is data fusion. Several studies have combined optical satellite images with Synthetic Aperture Radar (SAR) data to improve robustness under cloud cover and varying weather conditions. SAR provides structural and textural information, while optical images provide spectral information. Deep learning models have been used to fuse these modalities, leading to improved classification accuracy. However, such multimodal systems require more complex data processing pipelines and are computationally demanding.

Despite the significant progress achieved by deep learning models, many existing LULC classification systems still face practical limitations. Most research focuses primarily on improving classification accuracy, while issues such as model interpretability, user accessibility, and real-time deployment are often neglected. Deep learning models are typically treated as black boxes, making it difficult for users to understand why a particular land cover class was predicted. This lack of transparency reduces trust, especially in critical applications such as urban planning and environmental monitoring.

Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM have been introduced to address this issue. Grad-CAM generates heatmaps that highlight the regions of an image that contribute most to a model's prediction. In remote sensing, this helps users verify whether the model is focusing on meaningful land cover features such as vegetation, water bodies, or buildings. However, many existing LULC classification systems do not integrate XAI into their workflows.

Furthermore, most deep learning-based LULC classification studies are limited to offline experiments conducted in research environments. There is a lack of practical, user-friendly platforms that allow non-experts to upload satellite images and obtain classification results along with visual explanations. This creates a gap between academic research and real-world usage.

The proposed system addresses these limitations by combining high-performance deep learning models with Grad-CAM-based explainability and a Flask-based web application. By providing both accurate predictions and interpretable visualizations through an interactive platform, the system offers a more complete and practical solution for land use and land cover classification.

III. Existing System

The existing land use and land cover (LULC) classification systems primarily rely on traditional image processing and machine learning techniques. In most conventional approaches, satellite images are first preprocessed using radiometric and geometric corrections, after which handcrafted features such as spectral values, vegetation indices, texture measures, and shape descriptors are extracted. These features are then provided as input to classical classifiers such as Support Vector Machines (SVM), Random Forests, Decision Trees, or k-Nearest Neighbors to assign each pixel or image patch to a land cover class.

Although these methods have been widely used, they suffer from several limitations. One major drawback is the heavy dependence on manually designed features. Designing appropriate features requires expert knowledge and extensive experimentation, and the selected features often fail to capture the complex spatial patterns present in high-resolution satellite imagery. For example, urban areas, agricultural fields, and forests may have similar spectral characteristics but differ significantly in their spatial structures, which traditional feature-based methods struggle to model.

Another important limitation of existing systems is their sensitivity to noise, illumination variations, and seasonal changes. Satellite images captured at different times of the year or under different atmospheric conditions may exhibit significant variations in color and brightness. Traditional classifiers are not robust to these variations, leading to reduced accuracy and poor generalization across different regions and time periods.

Most conventional LULC classification systems also operate in a pixel-based or small patch-based manner, without considering the broader spatial context. As a result, they often produce noisy classification maps with scattered misclassifications, especially along boundaries between different land cover types. This makes it difficult to generate smooth and reliable land cover maps.

In addition, existing systems lack interpretability and transparency. The decision-making process of traditional classifiers is often difficult to visualize, and users cannot easily understand why a particular land cover class was assigned to a given region. This reduces trust in the system, particularly for applications that require high reliability, such as environmental monitoring and urban planning.

Furthermore, most existing LULC classification solutions are not designed for real-time or interactive use. They are typically implemented as offline tools used by experts, and there is no easy way for non-technical users to upload satellite images and obtain classification results through a simple interface. These limitations highlight the need for a more accurate, robust, interpretable, and user-friendly LULC classification system.

IV. Proposed System

The proposed system introduces a deep learning–based framework for automated and accurate land use and land cover classification using high-resolution satellite imagery. Unlike traditional methods that rely on handcrafted features, the proposed system employs convolutional neural networks (CNNs) to automatically learn hierarchical feature representations directly from raw image data. This enables the system to capture both low-level features such as edges and textures and high-level semantic patterns such as buildings, roads, vegetation, and water bodies.

To evaluate the effectiveness of different deep learning architectures, multiple CNN models are implemented and trained using the EuroSAT RGB dataset. These include ResNet50, ResNet18, DenseNet121, VGG19, ResNet152, and ConvNeXt-Tiny. Each of these architectures has unique design characteristics. ResNet models use residual connections to allow deeper networks to be trained efficiently. DenseNet promotes feature reuse through dense connectivity. VGG19 uses a deep and uniform architecture to capture detailed spatial patterns. ConvNeXt-Tiny represents a modern convolutional design that combines efficiency with strong representation learning.

Before training, satellite images are subjected to comprehensive preprocessing. Pixel values are normalized through rescaling, and data augmentation techniques such as geometric transformations and horizontal flipping are applied. These steps increase the diversity of training samples and help the models learn invariant features, improving their ability to generalize to unseen images. All images are reshaped to a uniform size to ensure compatibility with the CNN architectures.

During training, early stopping is employed to prevent overfitting and underfitting. This technique monitors the validation performance and stops training when no further improvement is observed, ensuring stable and reliable learning. After training, each model is evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of classification quality across different land cover classes.

To improve transparency and interpretability, the proposed system integrates Grad-CAM, an Explainable Artificial Intelligence (XAI) technique. Grad-CAM generates heatmaps that highlight the regions of the satellite image that contribute most to the model's prediction. This allows users to visually verify whether the model is focusing on meaningful land cover features, such as vegetation areas for forest classes or built-up regions for urban classes.

The best-performing model, ConvNeXt-Tiny, is selected based on experimental results and deployed in a Flask-based web application. The web platform includes SQLite-based user authentication to ensure secure access. Users can upload satellite images through the web interface, after which the system automatically preprocesses the image, performs classification, and displays both the predicted land cover category and the corresponding Grad-CAM heatmap. This end-to-end deployment transforms the deep learning model into a practical and accessible tool for real-world LULC analysis.

V. System Architecture

The proposed Land Use and Land Cover (LULC) classification system is designed using a modular and scalable client–server architecture that integrates deep learning, web technologies, and explainable AI. The architecture ensures efficient data flow, secure user interaction, and reliable model execution. It consists of five major components: the user interface, backend server, trained deep learning model, database, and visualization module.

The frontend layer provides an interactive web interface that allows users to register, log in, and upload satellite images for analysis. This interface is developed using HTML, CSS, JavaScript, and Bootstrap, enabling responsive design and easy interaction. Through this interface, users can select images from their local system and submit them for classification. The frontend also displays prediction results and Grad-CAM heatmaps generated by the backend.

The backend layer is implemented using the Flask web framework in Python. Flask acts as the core controller that receives requests from the frontend, manages user sessions, processes uploaded images, and communicates with the trained deep learning model. When an image is uploaded, the backend validates the file, applies necessary preprocessing steps, and passes the image to the model for inference. The predicted land cover class and Grad-CAM visualization are then returned to the frontend for display.

A centralized SQLite database is used to store user authentication data. This includes usernames, encrypted passwords, and session details. By using SQLite, the system ensures lightweight yet reliable storage, suitable for both local deployment and small-scale web hosting. Role-based access can be easily extended in the future to support multiple user types such as administrators and analysts.

The trained deep learning model, specifically ConvNeXt-Tiny, is stored as a serialized file. This model is loaded into memory by the backend at runtime to perform fast and efficient predictions. Because the model is pre-trained and optimized, it enables real-time classification of uploaded satellite images.

The Grad-CAM module forms the explainability layer of the system. After a prediction is generated, Grad-CAM computes gradients from the final convolutional layers of the CNN and produces a heatmap that highlights the most influential regions of the input image. This heatmap is overlaid on the original satellite image and returned to the user. This visual

explanation improves trust and transparency, allowing users to understand how the model makes its decisions.

Together, these components form a complete end-to-end architecture that supports secure user interaction, accurate LULC classification, and interpretable visual output.

VI. METHODS AND MATERIALS

The development of the proposed LULC classification system follows a structured methodology that includes dataset preparation, model training, evaluation, and deployment. The primary dataset used is the EuroSAT RGB dataset, which contains satellite images representing multiple land cover classes such as residential areas, industrial areas, forests, rivers, highways, agricultural fields, and water bodies. This dataset is widely used in remote sensing research and provides a realistic benchmark for evaluating classification performance.

In the first stage, dataset exploration is performed to understand the number of classes, image distribution, and resolution. Sample images are visualized to verify data quality and ensure that the dataset is properly organized. The dataset is divided into training, validation, and testing sets to allow fair and unbiased evaluation of the models.

Image preprocessing plays a critical role in improving model performance. Pixel values are rescaled to normalize intensity ranges, ensuring numerical stability during training. Data augmentation techniques such as rotation, shifting, zooming, and horizontal flipping are applied using an Image Data Generator. These operations increase the diversity of training samples and help the model learn invariant features, thereby reducing overfitting and improving generalization.

Multiple deep learning models are constructed using TensorFlow and Keras libraries. ResNet50, ResNet18, DenseNet121, VGG19, ResNet152, and ConvNeXt-Tiny are selected to provide a broad comparison of different architectural designs. Each model is initialized with pretrained weights and fine-tuned on the EuroSAT dataset. The networks are trained using backpropagation and stochastic gradient descent variants, with appropriate learning rates and batch sizes.

To prevent overfitting and underfitting, early stopping is applied during training. This technique monitors validation loss and stops training when performance no longer improves. This ensures that the models achieve stable and reliable convergence.

After training, model performance is evaluated using accuracy, precision, recall, and F1-score. These metrics provide a balanced view of classification quality, particularly in multi-class scenarios. Confusion matrices are also used to analyze class-wise performance.

Grad-CAM is applied to generate heatmaps for selected test images. These visualizations help interpret which regions of the satellite images influence the model's predictions, improving transparency and trust.

Finally, the best-performing model, ConvNeXt-Tiny, is integrated into a Flask-based web application. SQLite is used for user authentication, while the trained model handles real-time inference. This completes the transformation of the research model into a practical and usable system.

VII. Results

The proposed deep learning–based Land Use and Land Cover (LULC) classification system was evaluated using the EuroSAT RGB dataset to measure its effectiveness in accurately identifying different land cover categories. Multiple convolutional neural network architectures, including ResNet50, ResNet18, DenseNet121, VGG19, ResNet152, and ConvNeXt-Tiny, were trained and tested under the same preprocessing and training conditions to ensure a fair comparison.

The evaluation results show that all deep learning models significantly outperformed traditional feature-based classification methods, demonstrating the effectiveness of convolutional architectures in learning complex spatial and spectral patterns from satellite images. Among the tested models, ConvNeXt-Tiny achieved the highest performance across all evaluation metrics. It recorded an accuracy of 98.49%, a precision of 98.48%, a recall of 98.46%, and an F1-score of 98.47%. These values indicate that the model is both highly accurate and well-balanced in correctly identifying different land cover classes.

ResNet50 and ResNet152 also produced strong results, benefiting from their deep residual learning structures that enable stable gradient flow and effective feature extraction. DenseNet121 performed well due to its dense connectivity, which promotes feature reuse and enhances learning efficiency. VGG19, although computationally heavier, demonstrated good performance in capturing spatial details. However, ConvNeXt-Tiny outperformed all other models by combining modern convolutional design with efficient feature representation.

Grad-CAM visualizations were generated for a variety of test images. These heatmaps showed that the model focused on meaningful regions of the satellite images, such as vegetation areas for forest classes, water bodies for river and lake classes, and built-up regions for residential and industrial areas. This confirms that the model learned relevant spatial features rather than relying on irrelevant background patterns.

The integration of the trained ConvNeXt-Tiny model into the Flask web application was also tested. Users were able to upload satellite images and receive classification results along with Grad-CAM visual explanations in real time. The system handled multiple test inputs smoothly, demonstrating its practicality and stability for real-world usage.

VIII. Discussion

The experimental results highlight the strong potential of deep learning for accurate and reliable LULC classification.

Compared to traditional machine learning methods, CNN-based models are able to automatically learn complex spatial and contextual information from satellite imagery. This allows them to distinguish between visually similar land cover types that differ mainly in texture and spatial arrangement, such as urban areas and barren land or different types of vegetation.

Among the evaluated models, ConvNeXt-Tiny demonstrated superior performance due to its modern architectural design. Unlike older CNNs, ConvNeXt-Tiny uses optimized convolutional blocks and large receptive fields, allowing it to capture both local details and global context efficiently. This balance enables the model to achieve high accuracy while maintaining relatively low computational complexity, making it suitable for real-time and large-scale applications.

The inclusion of Grad-CAM plays an important role in improving trust and transparency. In many deep learning systems, users have little insight into how predictions are made. By visualizing the regions that influence classification decisions, Grad-CAM allows users to verify that the model is focusing on meaningful land cover features. This is especially important in remote sensing applications, where decisions may affect environmental policies, urban planning, and resource management.

The web-based deployment further enhances the practical value of the system. Instead of being limited to offline research experiments, the trained model is made accessible through a simple and intuitive interface. This allows researchers, students, and decision-makers to analyze satellite images without requiring deep technical knowledge of machine learning.

Despite its strong performance, the system has certain limitations. The current implementation uses only RGB satellite images, which may not capture all spectral information available in multi-spectral or hyperspectral data. In addition, while ConvNeXt-Tiny performs well, further improvements may be possible using larger datasets and more advanced architectures.

IX. Conclusion and Future Work

This paper presented a deep learning–based framework for automated and accurate land use and land cover classification using satellite imagery. By leveraging modern convolutional neural network architectures and comprehensive image preprocessing, the proposed system is able to achieve high classification accuracy across multiple land cover categories. Among the evaluated models, ConvNeXt-Tiny emerged as the best-performing architecture, achieving an accuracy of 98.49% along with high precision, recall, and F1-score values.

The integration of Grad-CAM provided valuable visual explanations, making the model's predictions more transparent and interpretable. The deployment of the trained model within a Flask-based web application with SQLite-based user authentication transformed the research model into a practical tool for real-time LULC analysis. Users can upload satellite images, obtain classification results, and view corresponding heatmaps, enabling informed and trustworthy decision-making.

Overall, the proposed system successfully addresses the limitations of traditional LULC classification methods by providing an accurate, scalable, and interpretable solution. It demonstrates how deep learning and explainable AI can be combined with web technologies to create powerful and user-friendly remote sensing applications.

In the future, the system can be extended in several ways. Multi-spectral and hyperspectral satellite data can be incorporated to capture richer spectral information, further improving classification accuracy. Temporal data from multi-date images can be used to perform land cover change detection and seasonal analysis. Advanced architectures such as vision transformers and attention-based models can be explored to enhance spatial–context modeling. Additionally, deploying the system on cloud platforms or edge devices can support large-scale, real-time environmental monitoring.

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